

Linear_regression

August 10, 2026

```
[1]: #imports
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
```

sample the data

```
[2]: data = {'Square_Feet': [1500, 1800, 2400, 3000, 3500, 4000, 4500, 500], 'Price':
    ↳ [400000, 450000, 500000, 600000, 650000,
    700000, 750000, 800000]}
df = pd.DataFrame(data)
```

split the data into training and testing sets

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[3]: X = df[['Square_Feet']] # Independent variable (feature)
y = df['Price'] # Dependent variable (target)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
    ↳ random_state=42)
```

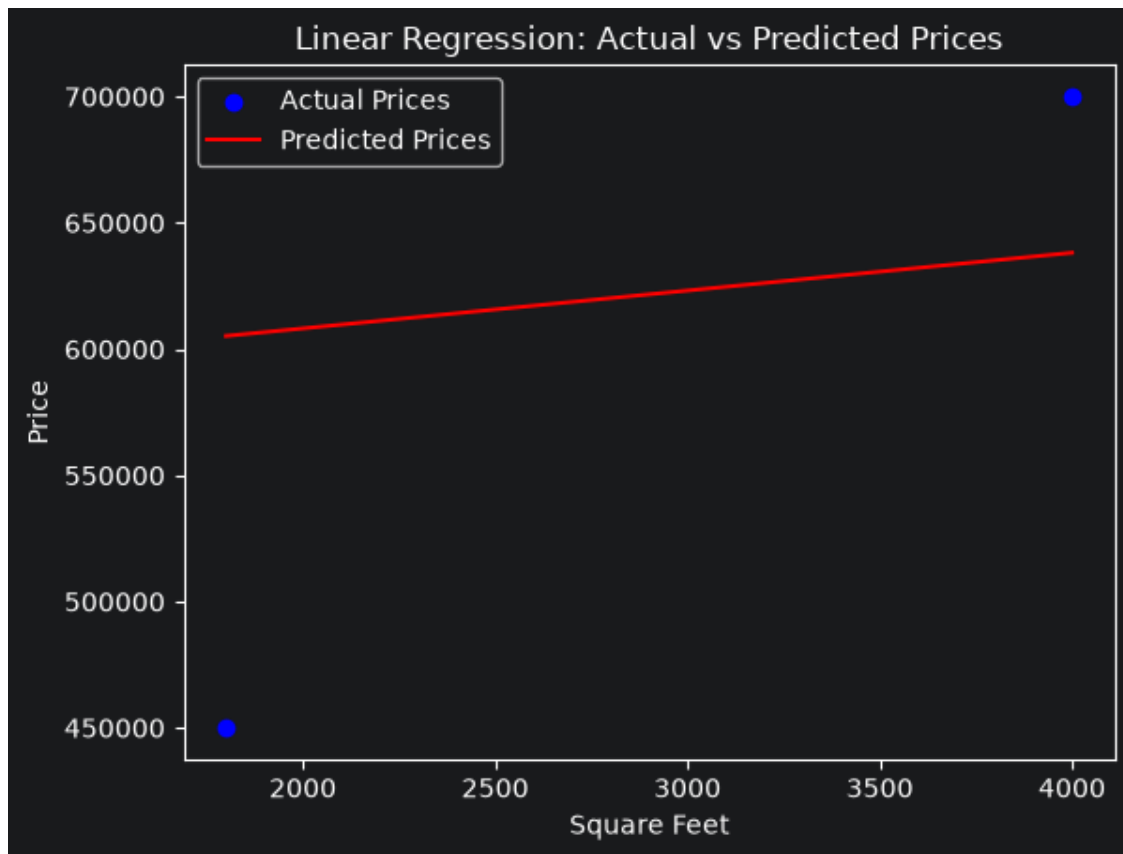
create the model and make predictions

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[4]: model = LinearRegression()
model.fit(X_train, y_train)

# Making predictions
y_pred = model.predict(X_test)
```

plot the results

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[5]: plt.scatter(X_test, y_test, color='blue', label='Actual Prices')
plt.plot(X_test, y_pred, color='red', label='Predicted Prices')
plt.title('Linear Regression: Actual vs Predicted Prices')
plt.xlabel('Square Feet')
plt.ylabel('Price')
plt.legend()
plt.show()
```



print the coefficients

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[6]: print(f'Intercept: {model.intercept_}')  
     print(f'Slope: {model.coef_[0]}')
```

Intercept: 578208.4690553746

Slope: 14.983713355048852